

VOROZHEYKA, Russian Federation

WMO: 292530

Lat: 58.456N

Lon: 89.469E

Elev: 134

StdP: 99.73

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.5	-37.6	-43.1	0.1	-40.4	-39.8	0.1	-36.9	7.4	-10.6	6.2	-10.3	0.0	90	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	28.3	19.6	26.5	18.5	24.9	17.8	21.4	26.1	20.1	24.6	18.9	23.2	1.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.7	23.8	18.4	13.5	22.5	17.2	12.5	21.2	63.2	26.3	58.5	24.5	54.4	23.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	4.4	3.8	-44.8	31.1	3.9	1.9	-47.6	32.5	-49.9	33.5	-52.1	34.6	-54.9	35.9		
(5)			-44.0	22.9	3.1	1.6	-46.3	24.1	-48.1	25.0	-49.8	25.9	-52.0	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-21.0	-16.3	-7.5	0.4	8.2	15.5	17.6	14.3	7.9	-0.1	-11.7	-17.9	(6)
(7)		DBStd	14.81	9.59	8.07	6.77	5.61	5.53	4.36	3.29	3.53	4.38	5.67	9.20	9.71	(7)
(8)		HDD10.0	4567	961	737	542	288	104	8	0	6	92	314	650	864	(8)
(9)		HDD18.3	7038	1219	971	800	537	317	102	53	131	313	572	900	1122	(9)
(10)		CDD10.0	629	0	0	0	1	48	174	236	141	29	1	0	0	(10)
(11)		CDD18.3	59	0	0	0	0	2	18	31	7	1	0	0	0	(11)
(12)		CDH23.3	699	0	0	0	1	42	264	299	85	8	0	0	0	(12)
(13)		CDH26.7	133	0	0	0	0	6	59	56	13	0	0	0	0	(13)
(14)	Wind	WSAvg	1.1	0.9	1.0	1.3	1.5	1.4	1.0	0.8	0.9	1.1	1.3	1.2	1.2	(14)
(15)	Precipitation	PrecAvg	501	28	22	22	30	44	53	51	58	57	50	48	42	(15)
(16)		PrecMax	692	73	44	49	65	105	133	111	135	134	97	82	84	(16)
(17)		PrecMin	344	1	9	3	8	12	0	7	19	14	15	15	5	(17)
(18)		PrecStd	85	15	9	10	15	25	27	24	32	27	18	15	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.5	0.7	8.7	17.8	27.2	30.9	30.5	28.5	24.5	16.0	5.4	0.7	(19)
(20)			MCWB	-2.5	-1.1	3.0	10.1	15.6	20.5	21.4	20.6	16.4	9.8	3.8	-0.7	(20)
(21)		2%	DB	-2.9	-1.3	5.9	14.5	24.1	28.3	28.4	25.7	20.8	12.3	3.1	-1.2	(21)
(22)			MCWB	-3.8	-2.6	1.4	7.7	13.9	19.1	20.5	18.1	14.5	7.7	1.5	-2.1	(22)
(23)		5%	DB	-4.9	-3.0	3.8	11.8	21.3	26.4	26.5	23.6	18.0	9.6	1.5	-3.3	(23)
(24)			MCWB	-5.8	-4.3	0.3	6.4	12.7	17.8	19.5	17.6	13.0	6.3	0.3	-4.1	(24)
(25)		10%	DB	-7.1	-5.1	1.9	9.1	18.3	24.4	24.9	21.4	15.2	7.1	0.1	-5.3	(25)
(26)			MCWB	-7.8	-6.1	-0.6	4.5	11.0	16.9	18.7	16.5	11.4	5.1	-0.9	-5.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.3	-0.8	4.4	11.0	16.5	22.3	23.5	21.8	17.8	10.3	4.3	-0.4	(27)
(28)			MCD B	-1.9	0.3	7.7	16.3	25.2	28.1	28.0	26.1	22.3	14.0	4.8	0.4	(28)
(29)		2%	WB	-3.7	-2.4	1.9	8.8	14.9	20.6	21.9	19.9	15.2	8.3	1.7	-2.1	(29)
(30)			MCD B	-3.0	-1.4	4.6	13.5	22.1	26.4	26.2	23.7	19.4	11.7	2.6	-1.3	(30)
(31)		5%	WB	-5.7	-4.2	0.7	6.6	13.4	19.1	20.7	18.5	13.7	6.6	0.3	-4.1	(31)
(32)			MCD B	-4.9	-3.2	3.3	11.3	19.7	24.4	24.9	22.0	17.1	8.9	1.3	-3.4	(32)
(33)		10%	WB	-7.8	-6.1	-0.5	4.8	11.7	17.8	19.5	17.1	12.0	5.2	-0.9	-5.9	(33)
(34)			MCD B	-7.0	-4.9	2.0	8.5	17.7	22.7	23.8	20.5	15.0	7.0	0.1	-5.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	10.4	11.3	10.3	11.0	11.7	11.2	10.5	9.2	6.7	7.2	7.7	(35)
(36)			MCDBR	7.2	8.3	12.5	14.0	15.8	15.8	13.9	14.3	14.6	11.0	5.5	7.5	(36)
(37)			MCWBR	6.8	7.4	9.2	9.3	8.7	7.8	7.2	7.9	8.9	7.3	5.1	7.3	(37)
(38)		5% WB	MCDBR	7.0	8.0	10.3	13.0	14.5	13.7	12.3	12.2	13.0	9.3	5.4	7.5	(38)
(39)			MCWBR	6.7	7.4	8.1	9.0	8.7	7.4	7.4	7.8	8.6	6.6	5.2	7.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.281	0.316	0.355	0.381	0.396	0.401	0.440	0.403	0.364	0.338	0.297	0.274	(40)
(41)		taud		1.968	1.970	1.946	2.055	2.247	2.321	2.198	2.292	2.360	2.246	2.014	1.919	(41)
(42)		Ebn at Noon		494	665	755	814	834	835	787	792	767	657	467	359	(42)
(43)		Edh at Noon		58	95	130	139	123	117	129	110	88	73	55	44	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.59	1.52	2.89	4.35	4.88	5.80	5.38	4.05	2.42	1.20	0.56	0.32	(44)
(45)		RadStd		0.07	0.12	0.23	0.39	0.51	0.61	0.36	0.38	0.36	0.13	0.11	0.06	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page